PRESENSE

SME AUTONOMOUS MAINTENANCE AGENT

From Reactive Break-Fix → Proactive Predictive Maintenance

Machine Learning · Model Context Protocol · Agentic AI

THE PROBLEM

ASEAN SMEs Are Flying Blind

Unplanned Downtime

Machines fail without warning, halting production lines for hours or days

RM 4,200/Hour Lost

Average cost of unplanned downtime for a Malaysian SME production line.

Reactive Break-Fix

70% of SMEs only service equipment after failure — no predictive strategy

Emergency Procurement

Panic-buying spare parts at 3–5× standard cost, disrupting cash flow.

THE SOLUTION

PreSense — Three Layers of Intelligence

01

ML ENGINE

- Random Forest Regressor
- NASA C-MAPSS benchmark
- RMSE: 19.00 (+54% vs baseline)
- SHAP Explainability

02

AGENTIC INTELLIGENCE

- Multi-factor Risk Score Engine
- PELT Change-Point Detection
- AUTO_DISPATCH / SCHEDULE / MONITOR
- Confidence-weighted decisions

03

MCP DISPATCH

- Real FastMCP Server
- check_inventory() tool
- schedule_technician() tool
- create_work_order() tool

TECHNICAL DEPTH

ML Model — NASA C-MAPSS Benchmark Results

MODEL PERFORMANCE

19.00 Official Test RMSE

+54% Improvement vs Baseline

1,207 S-Score (NASA metric)

100 Training Engines

KEY TECHNIQUES

Piecewise Capped RUL

Cap at 125 cycles — models real degradation physics, not linear decay

Rolling Mean Features

Window=3 smoothing on 14 sensors — eliminates noise without data leakage

SHAP Explainability

TreeExplainer shows exactly which sensors drive each prediction

PELT Change-Point

ruptures library detects regime shift mathematically — no hardcoded thresholds

AGENTIC AI + MCP

Real FastMCP Server + Decision Intelligence Engine

RISK SCORE ENGINE

$$\text{Risk Score} = \text{Failure Risk} \times \left(0.5 + \frac{\text{confidence}}{200} \right)$$

> 60 → AUTO_DISPATCH

> 30 → SCHEDULE

≤ 30 → MONITOR

@mcp.tool()

check_inventory()

Queries spare parts database by part name. Returns stock level, location, and unit cost.

@mcp.tool()

schedule_technician()

Finds available certified technician by specialisation. Returns name, contact, ETA.

@mcp.tool()

create_work_order()

Generates and dispatches work order with unit ID, cycle, RUL, and assignment details.

LIVE DEMO

PreSense Dashboard — Real-Time Demo

Choose machine from fleet

Stimulate time progression

AI detects regime shift

Risk score exceeds threshold

3 tools execute live

RM savings calculated

BUSINESS IMPACT

Market Potential & ASEAN ROI

70M+

ASEAN SMEs

Target addressable market

RM 4,200

Per Hour Lost

Avg unplanned downtime cost

54%

RMSE Improvement

vs baseline model

10x

ROI Potential

Per prevented failure/year

UN SDG ALIGNMENT

Goal 9: Industry, Innovation & Infrastructure

Target 9.4: Upgrade SME infrastructure with sustainable, resource-efficient and clean technologies across ASEAN.

COMPETITIVE EDGE

- No GPU required — runs on factory laptop
- SHAP explains every prediction to operators
- Real MCP server — not simulated
- PELT detects shifts mathematically
- Retrains on customer data in <2 mins

PRESENSE

Predict. Prevent. Protect.

ASEAN SME Predictive Maintenance · ML + MCP + Agentic AI

19.00	+54%	3 Tools
Test RMSE	Improvement	MCP Registered